



MARWADI UNIVERSITY

Faculty of **Engineering and Technology**

Department of Information and Communication Technology

B.Tech

SEM: 5

MID-SEM. EXAM: II

October: 2023

Subject: - Digital Signal and Image processing (**01CT0513**)

Date:-26-10-23

Total Marks:-30

Time: - 75 Minutes

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Do not write/sign/indication/tick mark anything other than Enroll No. at a specific place on the question paper.

Question: 1. LCCD equation of LTI System is as following:

[6]

$$y(n)=2y(n-1)+3x(n)+2x(n-1)$$

Find out

- (a) System Function (b) Impulse Response (b) Frequency Response (c) Magnitude Response
(d) Phase response (e) If input is $u(n)$ what will be output (f) If input is $\delta(n)$ what will be output

Question: 2.

[12]

- (a) Draw the Direct form I and Direct form II structure for a IIR filter having input output relationship $y(n)=x(n)+5x(n-1)+2y(n-1)+3y(n-2)$.

[6]

- (b) Find DTFT of following sequences (i) $(n+1)u(n)$ (ii) $e^{jkn} u(n)$

[6]

OR

- (b) Find circular convolution between $x(n)=\{1,3,2,3\}$ and $h(n)=\{1,1,1,1\}$.

[6]

Question: 3.

[12]

- (a) Find 8 point Decimation in time Fast Fourier transform $x(n)=\{1,1,1,1,0,0,2,0\}$

[8]

- (b) Differentiate Infinite Impulse Response and Finite Impulse Response Filter.

[4]

OR

- (a) Find 8 point Decimation in frequency Fast Fourier transform $x(n)=\{1,1,1,1,1,1,1,1\}$

[8]

- (b) What are the various methods for designing IIR filter. Explain any one technique.

[4]

---Best of Luck---